Beyond Planted Strings: Semantic Canaries, Canary Detectability, and Partial Unlearning in Anytime-Valid Unlearning Certificates

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Abstract

VOUCH certifies machine unlearning with paired ghost canaries: a fair coin decides which of two exchangeable records is trained and later unlearned, which makes the sign of their score difference an exact fair coin under exact unlearning, and betting e-processes turn these signs into an anytime-valid certificate on the committed cohort. The certificate was limited to random-string canaries and to a provider trusted not to treat canaries differently from other data. We extend it in three directions. First, we show that semantic twins, two incompatible natural-language facts about one fictitious entity drawn by a label-symmetric generator, keep the exact null for every score that compares the two facts under the same prompt, including belief scores over paraphrased questions never seen in training. Second, we prove that the cohort certificate transfers to the generator population at the cost of a McDiarmid term and to organic records at the cost of twice the provider's ability to distinguish canaries from organic records, which turns the honest-provider assumption into a quantitative statement. Every existing canary design is distinguished with advantage one; twins drawn from the organic record generator with one slot varied bring perplexity and n-gram detectors to chance. Third, on Pythia-410M and Pythia-1B, unlearning that targets the literal form removes it faster than the fact: the paraphrase advantage retains **0.21** more of its pre-unlearning value than the literal one ($p < 10^{-6}$ at both sizes), and a union audit over literal and paraphrase scores revokes up to **38%** of the runs that a literal-only audit would certify, while agreeing with it under no or strong unlearning. BASELINE$_S$ $ENTENCEWealsoboundthetimeatwhichthecohortcertificateisissuedthroughthegrowtho$ $populationspeed - up.$

Keywords: machine unlearning, anytime-valid inference, e-values, canaries, membership inference, large language models

1 Introduction

A user who asks a model provider to delete their data receives, at best, an assurance that an unlearning routine was run. Whether the routine removed what it was meant to remove is a statistical question, and VOUCH [1] gave it a statistical answer. Before fine-tuning, the provider plants *paired ghost canaries*: pairs of exchangeable records of which a fair coin admits one into training and, later, into the forget set, while the other is never trained on. Under exact unlearning the deployed model cannot depend on the coins, so the sign of the score difference between the two twins is a fair coin, exactly and at any sample size. Betting e-processes built on these signs give an anytime-valid (ε, α)-forgetting certificate, a revocation alarm, and a confidence sequence on the residual advantage, with no shadow models and no retrained reference.

The certificate VOUCH issues has a precise scope: it bounds the realised advantage of a declared class of scores on the committed cohort of canaries, under a provider that runs the same pipeline on canaries and on everything else. That scope leaves three questions open, and each has practical weight.

First, do canaries stand in for the data users delete? VOUCH's canaries are templated records with random secret strings. Users delete biographies, messages and facts. A certificate on random strings says nothing about natural-language facts unless something connects the two, and the connection has to survive a provider who might, deliberately or not, treat the two differently.

Second, what does a strategic provider need in order to pass the audit without unlearning? The honest-but-verifiable threat model of VOUCH assumes the provider does not special-case the cohort. The assumption is reasonable for an audit run with the provider's cooperation, but it is an assumption, and the original study measured that its own canary templates are easy to recognise. How much can a provider gain by recognising canaries, and what would it take to deny it that gain?

Third, does an audit of the trained form see knowledge that survives in other forms? Gradient-based unlearning can suppress the exact string it was given while the fact it encodes stays reachable through a paraphrase [2, 3]. A random-string secret has no paraphrase, so VOUCH's canaries cannot probe this failure. Semantic canaries can, provided the exact null survives the move to natural language.

Contributions.

This paper extends VOUCH along these three lines, with new theory and new experiments on Pythia models [4].

- **Semantic twins keep the exact null.** A twin pair that states two incompatible facts about one fictitious entity, drawn by a label-symmetric generator, keeps VOUCH's exact conditional null for every score that compares the two facts under the same prompt, including belief scores over paraphrased questions that never appear in training (Proposition 1). Calibrating by the base model removes the base model's prior between the two values without leaving the declared class.
- **A ladder from the cohort to organic records.** The cohort certificate transfers to the generator population with a McDiarmid term (Theorem 2) and from canaries to organic records with a term equal to twice the provider's ability to tell the two

apart (Theorem 3). The second result turns the honest-provider assumption into a quantitative statement: a provider who special-cases the cohort must operate a canary detector, and the gap it can open between canary and organic leakage is at most twice that detector's advantage (Corollary 4).

- **Detectability is measured and can be closed by construction.** Every canary design used so far, VOUCH's random-string records and naive semantic twins alike, is detected with advantage one by a character n-gram classifier. Twins drawn from the organic record generator itself, with a single slot varied, bring perplexity and n-gram detectors to chance level (Section 7.2).
- **Partial unlearning removes the trained form faster than the fact.** When unlearning targets the literal statements only, the literal advantage falls faster than the advantage on paraphrased questions never trained, by 0.21 in retention at both model sizes ($p < 10^{-6}$). Pythia-1B is harder to unlearn than Pythia-410M, and the strengths at which the two scores diverge move accordingly. A union audit over literal and paraphrase scores revokes in up to 38% more runs than a literal-only audit in this regime and agrees with it elsewhere (Section 7.4).
- **Comparison with membership-inference audits.** On the same runs, the standard membership-inference audit (an AUC test of members against a holdout set, with loss, Min-K%, zlib and reference-calibrated scores) is compared with VOUCH, including retraining as an exact-null control (Section 7.5).
- **When the cohort certificate arrives.** Under a uniformly random reveal order the betting edge of VOUCH's without-replacement process is deterministic (Lemma 5). A Freedman bound on the wealth process then gives a tail bound on the certification time that holds before refutation and exhibits the finite-population speed-up (Theorem 6); it is checked numerically and is conservative by a factor of two to four.

All code, results and the scripts that turn results into the tables of this paper are released with the VOUCH repository.

What is not claimed.

The organic-record result is conditional on the distinguishing advantage, which data can bound only from below; we report the strongest detectors we could build, not a proof that none better exists. All facts in the experiments are fictitious and the models have at most one billion parameters. These limits are discussed in Section 8.

Organisation.

Section 2 recalls VOUCH and fixes notation. Section 3 places the work in the literature. Sections 4–6 develop the theory: semantic twins, the ladder to organic records, and certification time. Section 7 reports the experiments and Section 8 discusses limitations.

2 Background: the VOUCH certificate

This section recalls the parts of VOUCH [1] that the extension builds on and fixes notation; proofs are in the original paper.

Pipeline and threat model.

A provider fine-tunes a model on a corpus, receives deletion requests, applies an unlearning routine to a forget set, and deploys the result M. An auditor (the verifier) wants evidence that the routine removed the influence of the forget set. The provider is *honest but verifiable*: it runs the committed pipeline on the committed data, and in particular processes the canaries exactly as it processes the rest of the forget set, but it is not trusted to report whether unlearning succeeded.

Paired ghost canaries.

Before fine-tuning, the verifier generates a cohort of m pairs with contents $C_i = (c_i^0, c_i^1)$ and draws independent fair coins $b_i \sim \mathrm{Bern}(\frac{1}{2})$. The in-twin $c_i^{b_i}$ is inserted into the training corpus and later into the forget set; the ghost twin $c_i^{1-b_i}$ is never trained on. The manifest of contents and coins is committed (VOUCH uses a Merkle commitment), and each coin is opened only when its pair is audited. All remaining randomness of training and unlearning is collected in ξ.

Scores, signs and the certified quantity.

A score $s(M, c)$ measures how much the model favours a record, for example the mean token log-probability of a secret after its prompt. A declared class $\mathcal{F}$ of scores is fixed before any query. For each pair and score the verifier computes the difference $D_i^{(s)} = s(M, c_i^{\mathrm{in}}) - s(M, c_i^{\mathrm{ghost}})$ and the sign $Z_i^{(s)} = \mathbf{1}[D_i^{(s)} > 0]$, ties split by a committed coin. The certified quantity is the realised cohort advantage

$$\Delta^{(s)} \;=\; \frac{2}{m} \sum_{i=1}^{m} Z_i^{(s)} \;-\; 1,$$

which is zero in expectation under exact unlearning and positive when the in-twins are still favoured. Negative values (the in-twin scores conspicuously below its partner) are leakage too, and VOUCH certifies both directions.

The exact null.

If M is conditionally independent of the coins given the contents, which holds for exact unlearning and for retraining from scratch, then for every score and every step t of a reveal order chosen independently of the signs, the sign of the pair opened at step t is a fair coin given everything opened before (VOUCH Theorem 1). No independence across pairs is needed: the signs share one model, but the coin governing each pair is fair given the past.

The certificate.

The verifier reveals the cohort in a uniformly random order π, drawn independently of the signs (VOUCH uses a public randomness beacon), and for each score and direction runs a betting wealth process $W_t = \prod_{u \le t} \big(1 + \lambda_u (p_0(u) - Z_{\pi(u)})\big)$ against

the *without-replacement boundary*

$$p_0(t) \;=\; \frac{mp_0 - S_{t-1}}{m - t + 1}, \qquad p_0 = \tfrac{1}{2} + \tfrac{\varepsilon}{2}, \tag{1}$$

where S_{t-1} is the number of ones revealed before step t. If the cohort advantage were at least ε, the conditional mean of the next sign would be at least $p_0(t)$, so W_t is a nonnegative supermartingale and Ville's inequality bounds the probability that it ever reaches $1/\alpha$. A two-sided (ε, α)-certificate is issued when every per-score, per-direction process has crossed $1/\alpha$, and then Pr[a false certificate is ever issued] $\leq \alpha$ under arbitrary dependence between pairs (VOUCH Theorem 2). If $p_0(t) > 1$ the null is infeasible and is refuted outright. A separate revocation process tests the exact null directly and raises an alarm (status REVOKED) when it accumulates evidence that the model still depends on the coins. A run that neither certifies nor revokes ends UNDETERMINED.

Transfer to the coin-averaged advantage.

The cohort advantage depends on the realised coins. If flipping one coin changes at most κ signs, McDiarmid's inequality [5] bounds the distance to the advantage averaged over the coins with the contents held fixed by $\kappa\sqrt{2\log(2/\alpha')/m}$ (VOUCH Proposition 3). Section 5 extends this transfer to the contents and then to organic records.

Notation for this paper.

We write Δ for the cohort advantage of a fixed score when the score is clear, $\mathcal{F}_t$ for the verifier's filtration after t reveals, and $\mathbb{E}$ for expectation. "Literal" refers to scores computed on the trained surface form of a canary, "paraphrase" to scores computed on questions never seen in training.

3 Related work

Unlearning methods and their evaluation.

Approximate unlearning for language models is dominated by gradient ascent on the forget set, negative preference optimisation [6] and its reference-free variant SimNPO [7], and representation misdirection [8]. They are evaluated on benchmarks such as TOFU [9] and MUSE [10] whose forget-quality and leakage metrics are fixed-sample statistics compared against a retrained model. A growing literature shows that these metrics overstate forgetting: forget-quality metrics track surface behaviour rather than retained knowledge [11], output matching hides degenerate models [12], benign relearning restores erased content [13], relearning attacks can be formalised and partly resisted [14], quantisation brings forgotten knowledge back [15], lightweight attacks recover unlearned capabilities [16], and current evaluations can be made to overstate or understate success [3]; see Liu et al. [17] and Xue et al. [18] for surveys. Closest to our partial-unlearning experiment, Li et al. [2] show that gradient-based unlearning shifts probability mass onto paraphrases of the targeted answer, which they call

squeezing, and propose an objective that accounts for the model's own beliefs. We study the measurement side of the same phenomenon: whether a certificate on the trained form can be misled, and what an audit needs to see it.

Membership inference as an audit.

Membership-inference attacks are the usual instrument of unlearning audits. Score-based attacks threshold the loss, its zlib-normalised version [19], or the mean of the least likely tokens, Min-K% Prob [20]; reference-based attacks calibrate the loss by a reference model, and LiRA [21] calibrates per example with shadow models. Hayes et al. [22] show that unlearning evaluations based on population-level attacks overstate privacy and that per-example calibration with shadow models is needed. Zhang et al. [23] establish that membership-inference scores cannot prove that a model was trained on given data unless the inclusion of that data was randomised, the principle behind VOUCH's fair coins. Our matched- and unmatched-holdout baselines (Section 7.5) show what happens to a membership-inference audit when that principle is not respected, and Theorem 3 states the same confound in the language of a certificate.

Verification of unlearning.

Thudi et al. [24] argue that approximate unlearning cannot be audited from weights alone and that auditable definitions are a prerequisite for any guarantee; Zhang et al. [25] show that a dishonest provider can defeat existing families of verification strategies while passing them; cryptographic proof-of-unlearning [26] attests that the committed computation ran, which is complementary to whether forgetting holds statistically. VOUCH [1] made statistical verification anytime-valid with paired canaries and betting e-processes [27–30], building on the secret-sharer methodology [31]. Privacy auditing with canaries [32] shares the planted design but certifies a privacy parameter of a training algorithm, not the outcome of one unlearning run. Yu et al. [33] show that retraining-based definitions of unlearning are not well posed for multi-stage pipelines, one reason VOUCH avoids a retrained reference. The special-casing provider that VOUCH's threat model excludes is the subject of Section 5: we do not remove the assumption, but we price it.

What is new here.

Relative to VOUCH, this paper (i) moves the canaries to natural language and proves the exact null survives, (ii) states and proves the transfer from canaries to organic records as a function of canary detectability, and measures detectability for several designs including one built to make it zero, (iii) tests the certificate against partial unlearning that suppresses the trained form, with two model sizes and a union audit, (iv) compares with membership-inference audits on the same runs, and (v) bounds the certification time through wealth growth rather than through refutation.

4 Semantic twins

VOUCH's secrets are random alphanumeric strings. A high-entropy string has no semantically equivalent rephrasing, so a score that reads the secret under paraphrased carrier sentences finds the same advantage as the literal score, and the squeezing effect of Li et al. [2] cannot be observed; the original paper reports exactly this and names natural-language secrets, exchangeable as propositions rather than as strings, as the next step [1]. This section defines such canaries and shows that the exact null survives.

4.1 The generator

A semantic twin pair names a fresh fictitious entity e and an attribute a (for example birthplace or profession) and states two incompatible values of it: c^0 says that e has value v, c^1 that e has value $v' \neq v$. The pair is drawn by a generator controlled by the verifier.

Assumption 1 (Generator) The pair contents $C_1, \ldots, C_m$ are independent draws from a generator distribution P on ordered pairs that is *label-symmetric*: $(c^0, c^1) \sim P$ implies $(c^1, c^0) \sim P$. The coins b_i are independent Bern($\frac{1}{2}$) draws, independent of the contents and of ξ, and ξ is independent of the contents.

Both properties are under the verifier's control, because they describe its own sampler rather than the data or the model. Our generator draws the two values as an unordered pair from the attribute's value set and swaps them with probability one half, which makes it label-symmetric by construction; the coins come from a separate random stream. The in-twin is trained in one or more *training forms*, for example the literal statement "e was born in v" and one question–answer form.

4.2 Scores

Each score compares the two values under the same prompt, so that both twins of a pair are scored by the same function of the model:

- *literal*: the mean token log-probability of the value after the literal prefix ("e was born in"), $s_{\mathrm{lit}}(M, c^k) = \overline{\log P_M}(v_k \mid \text{prefix})$;
- *belief*: the same quantity averaged over a committed set Q of paraphrased questions ("Where was e born?", "What is the hometown of e?"), none of which appears in training;
- *belief_max*: the maximum over Q instead of the mean, the semantic analogue of VOUCH's s_{para};
- the *calibrated* version of each, which subtracts the same score computed under the frozen base model M_0: $s^{\mathrm{cal}}(M, c) = s(M, c) - s(M_0, c)$.

Calibration removes the base model's prior preference between the two values (a base model may find "born in Paris" more likely than "born in Porto" for any name). It keeps the score in the declared class because M_0 is fixed before the coins are drawn.

In the experiments we refer to the literal score as *literal* and to the calibrated belief scores as *paraphrase*.

4.3 The exact null

Proposition 1 (Exact null for semantic twins) *Under Assumption 1, if the deployed model is conditionally independent of the coins given the contents, then for every score s of the form above and every step t of a reveal order chosen independently of the signs, $Z^{(s)}_{\pi(t)} \mid \mathcal{F}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$ exactly.*

Proof The proof of VOUCH Theorem 1 uses only that, given the unordered pair opened at step t, the two numbers $s(M, c^0)$ and $s(M, c^1)$ are determined, and that the coin deciding which of them belongs to the in-twin is fair given everything opened before and independent of the model given the contents. Neither the semantics of the records nor the form of s enters, provided $s(M, \cdot)$ is a function of the model and the record alone. Each score above is such a function: both twins share the prefix or the question set and differ only in the value evaluated, and the calibrated score subtracts a function of M_0 and the record, where M_0 does not depend on the coins. □

Two remarks delimit what the proposition gives. First, *semantics costs power, not validity.* A base-model prior between the two values enters both arms symmetrically and leaves the null exact, but it adds variance to the score differences; calibration removes most of it. Second, the null does not need label symmetry: the coin, not the order of the pair, decides which twin is trained. Label symmetry matters for what the certificate says about records outside the cohort. It makes the two components c^0 and c^1 of a pair identically distributed, so every canary record is a draw from a single record distribution, whichever position it occupies and whichever twin the coin trains, and it is this distribution that Section 5 compares with the records users actually delete.

5 From the cohort to organic records

The VOUCH certificate concerns the realised advantage Δ on one committed cohort. This section climbs a ladder of three targets: the cohort, the generator population, and organic records, the data users actually delete. Each step costs one term in the tolerance, and each term has its own cause.

5.1 From the cohort to the generator population

VOUCH Proposition 3 averages over the coins with the contents fixed. The verifier also controls the contents, through Assumption 1, so the same argument reaches them.

Assumption 2 (Bounded influence) For fixed ξ, changing a single coin b_i changes at most κ of the m signs, and replacing the contents C_i of a single pair changes at most ℓ of them.

Under exact unlearning $\kappa = \ell = 1$: the model carries no trace of the forget set, so a pair's coin or contents can affect only that pair's own sign. Under partial unlearning

κ and ℓ measure how far one canary's training reaches into the others' scores; they are small for canaries about distinct fictitious entities.

Theorem 2 (Transfer to the generator population) *Let $\Delta_\xi = \mathbb{E}_{b,C}[\Delta \mid \xi]$ be the expected advantage of this pipeline run over fresh cohorts from P. Under Assumptions 1 and 2, for any $\alpha' \in (0,1)$, with probability at least $1-\alpha'$ over (b, C),*

$$|\Delta - \Delta_\xi| \;\le\; u_m(\alpha') \;:=\; \sqrt{\frac{2(\kappa^2+\ell^2)\log(2/\alpha')}{m}}.$$

Consequently a two-sided (ε, α) cohort certificate implies $|\Delta_\xi| < \varepsilon + u_m(\alpha')$ with probability at least $1-\alpha-\alpha'$.

Proof Given ξ, Δ is a function of the $2m$ independent coordinates $(b_1, \ldots, b_m, C_1, \ldots, C_m)$. By Assumption 2, changing one coin changes Δ by at most $2\kappa/m$ and changing one pair's contents by at most $2\ell/m$. McDiarmid's inequality [5] gives $\Pr[|\Delta - \Delta_\xi| \ge u] \le 2\exp(-2u^2/\sum_k c_k^2)$ with $\sum_k c_k^2 = 4(\kappa^2+\ell^2)/m$, that is $2\exp\big(-u^2 m/(2(\kappa^2+\ell^2))\big)$; equating it to α' gives u_m. The consequence follows from a union bound with the certificate's error probability. □

Δ_ξ averages over cohorts but conditions on the training randomness ξ: it is the advantage of *this* deployed pipeline run on the generator population, the relevant target for an auditor of a deployed model. Averaging over ξ would need several independent training runs and is not attempted. At $m = 512$, $\alpha' = 0.05$ and $\kappa = \ell = 1$ the inflation is $u_m = 0.17$, against 0.12 for the coin-only transfer.

5.2 From canaries to organic records

Let P_o be a distribution of *organic* record pairs: two records of the kind a user deletes, with a hypothetical fair coin choosing which one was trained on. For pair contents c, let

$$h_\xi(c) \;=\; \mathbb{E}\big[\, 2Z_i - 1 \;\big|\; C_i = c,\; \xi \,\big]$$

be the expected sign contribution of a pair with contents c placed in the cohort, the expectation being over the coins and the other pairs' contents. Then $\Delta_\xi = \mathbb{E}_{c\sim P}[h_\xi(c)]$, and we define the *organic advantage* $\Delta_\xi^o = \mathbb{E}_{c\sim P_o}[h_\xi(c)]$: the advantage the same pipeline run would show on organic pairs planted in the same way. This is the standard notion of membership leakage under randomised inclusion, evaluated on the data users delete; because organic records were not in fact included by a coin, it is a counterfactual quantity of the pipeline, not an observed rate.

Definition 1 (Distinguishing advantage) For a class $\mathcal{H}$ of functions from pair contents to $[0,1]$, $\gamma_{\mathcal{H}}(P, P_o) = \sup_{\phi\in\mathcal{H}} \big|\mathbb{E}_P\phi - \mathbb{E}_{P_o}\phi\big|$. When $\mathcal{H}$ contains all measurable functions, $\gamma_{\mathcal{H}}$ is the total variation distance between P and P_o.

Theorem 3 (Special-casing requires detection) *Let $\phi_\xi = (1+h_\xi)/2$, which takes values in $[0,1]$. Then*

$$|\Delta_\xi^o - \Delta_\xi| \;=\; 2\,\big|\mathbb{E}_P\phi_\xi - \mathbb{E}_{P_o}\phi_\xi\big| \;\le\; 2\,\gamma_{\mathcal{H}}(P, P_o)$$

for every class $\mathcal{H}$ that contains ϕ_ξ. Equivalently, if the pipeline treats canaries and organic records differently enough to open a gap g between their advantages, then ϕ_ξ is a test that tells canary pairs from organic pairs with advantage at least $g/2$.

Proof By linearity, $\Delta_\xi - \Delta_\xi^o = \mathbb{E}_P h_\xi - \mathbb{E}_{P_o} h_\xi = 2(\mathbb{E}_P \phi_\xi - \mathbb{E}_{P_o} \phi_\xi)$, and the right-hand side is at most $2\gamma_\mathcal{H}$ in absolute value whenever $\phi_\xi \in \mathcal{H}$. □

The proof is one line; the content is in the reading. The function ϕ_ξ is computable by anyone who can run the provider's pipeline on a pair and score both twins, so the relevant class $\mathcal{H}$ is "tests the provider can implement". The theorem thus turns VOUCH's honest-but-verifiable assumption into a quantitative one: a provider that special-cases the cohort, unlearning canaries while leaving organic records untouched, must in effect operate a canary detector, and the gap it can open is at most twice that detector's advantage. Three consequences follow.

1. If the manifest is committed but its contents are hidden until after deployment, the provider can recognise canaries only from their contents, which is what $\gamma_\mathcal{H}$ measures.
2. An auditor cannot certify $\gamma_\mathcal{H}$ from data. Every detector we train gives a *lower* bound; none gives an upper bound. We therefore report the strongest detectors we could build and state the organic claim conditionally on γ.
3. Making P close to P_o shrinks γ against every detector at once. In the limit where each canary record is drawn from the organic record distribution itself, $\gamma = 0$ at the level of single records.

Corollary 4 (The ladder) *Under the assumptions of VOUCH Theorem 2 and of Theorems 2 and 3, if a two-sided (ε, α) cohort certificate is issued, then with probability at least $1-\alpha-\alpha'$,*

$$|\Delta_\xi^o| \;<\; \varepsilon \;+\; u_m(\alpha') \;+\; 2\gamma_\mathcal{H}(P, P_o).$$

The three terms have separate causes: the tolerance the auditor chose, the size of the cohort, and how far the canaries are from the data they stand in for. Only the third depends on the canary design, and Section 7.2 measures it.

5.3 Organic-matched twins

The third consequence above suggests a design. Let the organic records be produced by a generator G_o, for example a set of sentence templates with slot distributions, and draw each canary pair as follows: sample one record $r \sim G_o$, choose one of its slots (fixed per template), and form the pair by filling that slot with an unordered pair of values drawn from the slot's distribution and swapped by a fair coin. Each twin is then a record of G_o with one slot redrawn, so under a slot distribution that is exchangeable over its values each twin is distributed exactly as an organic record. We call these *organic-matched twins*. Two details matter. Names are drawn as in organic data (no distinguishing middle initial), and each canary appears once, since a record

repeated verbatim is itself a signal that organic data rarely gives. The price is power: a record that looks organic is no easier to memorise than an organic record.

Matching is easy here because we wrote the organic generator. For a deployed model the organic distribution is the users' own data, and matching it is the hard part of the problem; Section 8 returns to this.

6 When the cohort certificate arrives

VOUCH bounds the certification time of the cohort certificate through refutation (VOUCH Theorem 5): once enough zeros have been revealed, the null becomes infeasible and is rejected at any level. Refutation is exact but late, since it waits until the unrevealed remainder cannot carry the null. In practice the wealth process crosses $1/\alpha$ much earlier, by ordinary growth. This section bounds that time. The key observation is that under a uniform reveal order the betting edge of the without-replacement process is deterministic.

Lemma 5 (Deterministic edge) *Consider the upper certificate arm with $p_0 = \frac{1}{2} + \frac{\varepsilon}{2}$. Let the cohort carry j ones, $\hat{p} = j/m < p_0$, and $\delta = p_0 - \hat{p}$. Under a uniform reveal order, the boundary $p_0(t)$ of (1) and the conditional mean $r_t = \mathbb{E}[Z_{\pi(t)} \mid \mathcal{F}_{t-1}]$ satisfy, for every t and every history,*

$$g_t \;:=\; p_0(t) - r_t \;=\; \frac{m\delta}{m - t + 1}.$$

Proof With S_{t-1} ones revealed, $p_0(t) = (mp_0 - S_{t-1})/(m - t + 1)$ and, since the unrevealed remainder is a uniformly ordered set containing $j - S_{t-1}$ ones, $r_t = (j - S_{t-1})/(m - t + 1)$. Subtracting cancels S_{t-1}. □

The edge grows as the cohort is exhausted. On an i.i.d. stream with the same mean it would stay equal to δ; the growth is the finite-population advantage of the cohort target.

Theorem 6 (Certification time through wealth growth) *Run the upper arm with a uniform mixture of K constant stakes $\Lambda = \{\lambda_1, \dots, \lambda_K\} \subset (0, \frac{1}{2}]$, refuting the null outright if $p_0(t) > 1$. Let H_n denote the harmonic numbers,*

$$A_t = \sum_{u \le t} g_u = m\delta\,(H_m - H_{m-t}), \qquad G_t = \sum_{u \le t} g_u^2 = m^2\delta^2 \sum_{k=m-t+1}^{m} k^{-2},$$

and for $\lambda \in \Lambda$

$$\Psi_t(\lambda) = \lambda A_t - \frac{\lambda^2(t/4 + G_t)}{2(1-\lambda)^2}, \qquad V_t(\lambda) = \frac{\lambda^2(t/4 + G_t)}{(1-\lambda)^2}, \qquad R_\lambda = \log\frac{1+\lambda}{1-\lambda}.$$

Then the stopping time τ of the arm (wealth reaching $1/\alpha$, or refutation) satisfies, for every $t < m$,

$$\Pr[\tau > t] \;\le\; \min_{\lambda \in \Lambda} \exp\Big(-\frac{x_t(\lambda)^2}{2\big(V_t(\lambda) + R_\lambda x_t(\lambda)/3\big)}\Big), \qquad x_t(\lambda) = \big(\Psi_t(\lambda) - \log(K/\alpha)\big)_+.$$

Proof Fix $\lambda \in \Lambda$ and let $X_u = \log\bigl(1+\lambda(p_0(u) - Z_{\pi(u)})\bigr)$, so that $\log W_t^\lambda = \sum_{u \le t} X_u$. Before refutation $p_0(u) \in [0,1]$, so $y_u = \lambda(p_0(u) - Z_{\pi(u)}) \in [-\lambda, \lambda]$. By Taylor's theorem with remainder, $\log(1+y) \ge y - y^2/(2(1-\lambda)^2)$ for $y \ge -\lambda$. Hence $\mathbb{E}[X_u \mid \mathcal{F}_{u-1}] \ge \lambda g_u - \lambda^2\,\mathbb{E}[(p_0(u) - Z_{\pi(u)})^2 \mid \mathcal{F}_{u-1}]/(2(1-\lambda)^2)$, and $\mathbb{E}[(p_0(u) - Z_{\pi(u)})^2 \mid \mathcal{F}_{u-1}] = r_u(1-r_u) + g_u^2 \le \frac{1}{4} + g_u^2$. Summing over $u \le t$, the compensator of $\log W_t^\lambda$ is at least $\Psi_t(\lambda)$, a deterministic quantity by Lemma 5. Further, $|\log(1+y)| \le |y|/(1-\lambda)$ for $|y| \le \lambda$, so the conditional variances of the X_u sum to at most $V_t(\lambda)$, and the centred increments are bounded below by $-R_\lambda$. After refutation, replace X_u by its compensator lower bound, so that the centred sequence remains a martingale difference sequence; this changes nothing on the event that refutation has not occurred by t. Freedman's inequality [34], applied to the negated increments, gives $\Pr[\log W_t^\lambda < \Psi_t(\lambda) - x] \le \exp\bigl(-x^2/(2(V_t(\lambda) + R_\lambda x/3))\bigr)$. Finally, the mixture wealth is at least W_t^λ/K, so on the event of no refutation by t we have $\{\tau > t\} \subseteq \{\log W_t^\lambda < \log(K/\alpha)\}$. Taking $x = x_t(\lambda)$ and minimising over Λ completes the proof. □

Reading the bound.

Since $1/(m-u+1) \ge 1/m$, $A_t \ge \delta t$, with equality only in the limit of an infinite cohort; the factor $\rho_t = m(H_m - H_{m-t})/t \approx 1 + t/(2m)$ is the finite-population speed-up. Ignoring G_t and the factor $(1-\lambda)^{-2}$ and optimising λ, the drift is about $2\delta^2 t\rho_t^2$, so the time scale is $t^* \approx \log(K/\alpha)/(2\delta^2\rho_{t^*}^2)$, against $\log(1/\alpha)/\mathrm{KL} \approx \log(1/\alpha)/(2\delta^2)$ for an i.i.d. stream. The original study found its cohort certificate faster than the super-population one at every tolerance and attributed this to the cap at cohort exhaustion, where the claim resolves deterministically. The bound shows that the speed-up starts well before exhaustion, through the edge that grows as the cohort runs out. The lower arm is the mirror statement with $\delta' = \hat{p} - (1 - p_0)$, and the two-sided certificate needs both arms. The theorem covers a constant-stake mixture. VOUCH's engine uses adaptive experts (online Newton step, Krichevsky–Trofimov); adding the constant-stake grid to the engine's experts inherits the bound at the cost of log of the number of experts.

Numerical check.

We compare the bound with the empirical tail of the same mixture bettor ($K = 10$ stakes, $0.05, 0.10, \ldots, 0.50$) over 2,000 uniformly random orders, at $\hat{p} = \frac{1}{2}$ and $(m, \varepsilon) \in \{(512, 0.2), (1024, 0.2), (1024, 0.1), (2048, 0.1)\}$ (Table 1). The bound is never violated at any of the 32 checked times per setting. It is conservative: the first time at which it falls below one half is 2.2 to 3.8 times the empirical median. The same bettor against a fixed boundary on an i.i.d. stream with the same mean is slower in every setting.

7 Experiments

The experiments ask four questions. How detectable are canaries, and can detectability be closed by design (Theorem 3)? Does an audit with semantic twins separate memorising from unlearned models on a real language model (Proposition 1)? Under partial unlearning, does the trained form disappear before the fact, and does auditing paraphrases change verdicts? How do membership-inference audits behave on the same runs? Every number below is produced by a script under `extension/experiments/`, stored under `extension/results/`, and turned into the tables of this paper by

Table 1 Certification time of the upper arm, $\hat{p} = \frac{1}{2}$, $\alpha = 0.05$, 2,000 uniformly random reveal orders per setting. "Bound $\leq \frac{1}{2}$" is the first checked time at which Theorem 6 bounds $\Pr[\tau > t]$ by one half; "fixed boundary" is the same bettor on an i.i.d. stream with the same mean.

m	ε	median τ	bound $\leq \frac{1}{2}$	ratio	fixed boundary	violations
512	0.2	119	346	2.9	142	0/32
1024	0.2	130	490	3.8	138	0/32
1024	0.1	370	810	2.2	527	0/32
2048	0.1	427	1226	2.9	532	0/32

`extension/manuscript/make_tables.py`. A test suite of eleven tests checks the generators and the exact null without model weights.

7.1 Setup

Data.

The organic records are fictitious biographies drawn from six sentence templates (birthplace and profession; study city; publication city; years in a city before a profession; a city guild prize; a workshop near a city) over fictitious first and last names, cities and professions. These are the keep and forget biographies of the VOUCH experiments. Each run uses 2,000 keep records, 200 organic forget records and a cohort of $m = 384$ semantic twin pairs over five attributes (birthplace, profession, instrument, pet, colour). In-twins are trained in the literal form and one question–answer form and are repeated 1, 2, 4 or 8 times by stratum. Belief scores use four paraphrased questions per attribute, none of which is trained.

Models and training.

The main model is Pythia-410M and the second size is Pythia-1B [4]; GPT-2 is used for a pilot and for detector features. A LoRA adapter [35] of rank 16 on all linear layers is fine-tuned for 600 steps (batch 8, learning rate $2 \cdot 10^{-4}$, blocks of 96 tokens). Gradient ascent (GA) and NPO [6] act on a copy of the fine-tuned adapter with batch 4 and learning rate 10^{-4}; NPO uses a retain term on the keep data and the fine-tuned adapter as its reference. Retraining is a fresh adapter trained on the keep data only, which satisfies the exact null; no unlearning is the positive control. All runs were done on single NVIDIA T4 GPUs.

Audit.

Verdicts use the VOUCH engine at $\alpha = 0.05$ with a reveal order drawn from local entropy for the reported runs and from committed seeds for the offline re-analyses. We report the realised advantage $\hat{\Delta}$ of the literal score and of the calibrated belief score ("paraphrase"). Recall that REVOKED rejects the exact null, some dependence on the coins, at any ε, while ISSUED certifies $|\Delta| < \varepsilon$.

Table 2 Detecting canaries among organic records (400 canaries, 400 organic records, five-fold out-of-fold scores). A detector's advantage is a lower bound on the distinguishing advantage γ of Theorem 3. The null reference for the advantage is 0.095 (95th percentile) and 0.115 (99th).

	Perplexity features		Character n-grams		
Canary design	AUC	advantage	AUC	advantage	$\gamma \geq$
VOUCH random-string QA	0.998	0.973	1.000	1.000	1.000
semantic twin, literal	0.998	0.978	1.000	1.000	1.000
semantic twin, question–answer	0.892	0.605	1.000	1.000	1.000
organic-matched twin	0.521	0.052	0.442	0.102	0.102

7.2 Canary detectability

Theorem 3 bounds the gap between canary and organic advantage by twice the distinguishing advantage γ. We train two detectors to tell 400 canary in-twins from 400 organic records, both scored out of fold over five folds: a logistic regression on GPT-2 token-loss features (mean, maximum, median and spread of the token losses, and length), a trained version of the zero-knowledge perplexity filter of the original study, and a logistic regression on character 2–4-grams, which requires labelled canaries. A detector's advantage $\max_\tau |\mathrm{TPR} - \mathrm{FPR}|$ is a lower bound on γ. For reference, a detector whose scores are independent of the label exceeds 0.095 in 5% and 0.115 in 1% of 4,000 simulations at this sample size.

Table 2 compares four designs. VOUCH's random-string records and both forms of naive semantic twins are detected with advantage one by the n-gram classifier, and the first two also by the zero-knowledge perplexity filter. Random strings, a middle initial in every canary name and attributes that never occur in organic data give the cohort away. For these designs Corollary 4 says nothing about organic records. Organic-matched twins (Section 5.3) bring both detectors to chance: the perplexity detector's advantage, 0.052, is below the 95th percentile of the null reference, and the n-gram detector's, 0.102, lies between the 95th and the 99th; its AUC is below one half, so the classifier does not separate the classes in the direction it was trained for, and the advantage comes from the absolute value in its definition. These are lower bounds on γ against two detector families, not a proof that no detector exists, which is the conditional nature of Theorem 3. The original study found that even an entropy-diluted template that defeats the perplexity filter is still recovered by a trained n-gram classifier; matching the organic generator removes that signature too.

7.3 Auditing with semantic twins

Table 3 audits Pythia-410M when the forget set contains every form the in-twins were trained in.

Paraphrased questions that were never trained carry about two thirds of the literal advantage, and the audit separates the positive from the negative control in every seed. A pilot with GPT-2 (128 pairs, 400 training steps) did not: its positive control

Table 3 Semantic-twin audit on Pythia-410M, full forget set ($m = 384$, three seeds). Realised advantages per seed and the two-sided verdict over all scores (I issued, R revoked, U undetermined).

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	$\varepsilon = 0.2$	$\varepsilon = 0.1$
no unlearning	+0.29, +0.32, +0.46	+0.20, +0.27, +0.26	R R R	R R R
GA (100 steps)	+0.03, +0.07, −0.05	+0.05, +0.05, −0.07	I I I	I I I
NPO (250 steps)	+0.03, −0.09, −0.04	+0.07, −0.07, −0.03	I I I	I I I
retrain	−0.03, −0.05, +0.03	+0.01, +0.11, +0.08	I I I	I U I

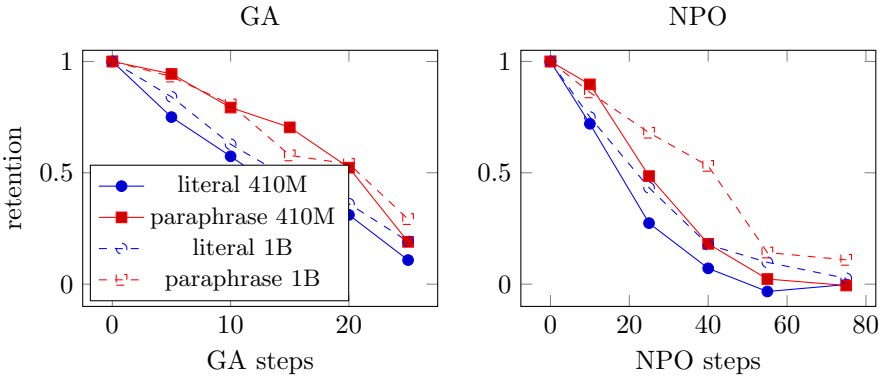


Fig. 1 Retention of the literal and the paraphrase advantage under literal-only unlearning, mean over eight seeds ($m = 384$). Solid: Pythia-410M; dashed: Pythia-1B.

reached advantages of +0.25 (literal) and +0.17 (paraphrase) and was not revoked. Semantic twins need a model that generalises a stated fact to its paraphrases, which the smaller model barely does. With the full forget set, GA and NPO remove both advantages and nothing distinguishes the two scores.

7.4 Partial unlearning

The case a literal audit cannot see is unlearning that removes the trained surface form but not the fact. To provoke it we restrict the forget set to the literal statements, so the question–answer form stays out of it, and sweep the strength of unlearning: GA for 5 to 25 steps and NPO for 10 to 75 steps, with eight seeds per model; three of the Pythia-410M seeds also run GA for 50 and 100 steps and NPO for 250. The *retention* of a score is its advantage after unlearning divided by the same seed's advantage before unlearning.

Figure 1 and Table 4 support three findings.

The literal form is unlearned faster than the fact, at both sizes. At every strength where both advantages are still clearly non-zero, paraphrase retention exceeds literal retention on average. Pooled over GA 10–20 and NPO 25–40 (forty seed-level cells per model), the mean difference is 0.21 ± 0.03 for Pythia-410M (positive in 34 of 40

Table 4 Literal-only unlearning on Pythia-410M (top) and Pythia-1B (bottom), eight seeds each. Advantages are means ± standard errors; p is the Wilcoxon signed-rank test of paraphrase minus literal retention (smallest attainable two-sided value 0.008). The last two columns count seeds in which the single-score audits disagree at $\varepsilon = 0.2$: A, literal issued and paraphrase revoked; B, the reverse.

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	retention lit. / para.	p	A	B
none	$+0.363 \pm 0.026$	$+0.219 \pm 0.016$	1.00 / 1.00	–	0	1
GA 5	$+0.275 \pm 0.028$	$+0.207 \pm 0.019$	0.75 / 0.94	0.039	0	1
GA 10	$+0.210 \pm 0.024$	$+0.175 \pm 0.020$	0.57 / 0.79	0.039	1	1
GA 15	$+0.143 \pm 0.016$	$+0.157 \pm 0.020$	0.39 / 0.70	0.008	1	1
GA 20	$+0.113 \pm 0.023$	$+0.120 \pm 0.019$	0.31 / 0.52	0.008	2	2
GA 25	$+0.036 \pm 0.015$	$+0.038 \pm 0.016$	0.11 / 0.19	0.250	0	1
NPO 10	$+0.259 \pm 0.019$	$+0.195 \pm 0.020$	0.72 / 0.90	0.039	0	0
NPO 25	$+0.097 \pm 0.015$	$+0.109 \pm 0.022$	0.27 / 0.49	0.055	3	1
NPO 40	$+0.020 \pm 0.027$	$+0.046 \pm 0.026$	0.07 / 0.18	0.109	2	0
NPO 55	-0.013 ± 0.014	$+0.007 \pm 0.028$	-0.03 / 0.02	0.742	1	0
NPO 75	-0.006 ± 0.017	$+0.005 \pm 0.021$	-0.00 / -0.01	0.844	0	0

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	retention lit. / para.	p	A	B
none	$+0.368 \pm 0.034$	$+0.209 \pm 0.031$	1.00 / 1.00	–	0	1
GA 5	$+0.314 \pm 0.037$	$+0.199 \pm 0.033$	0.84 / 0.93	0.016	0	4
GA 10	$+0.241 \pm 0.038$	$+0.177 \pm 0.035$	0.63 / 0.81	0.055	0	2
GA 15	$+0.186 \pm 0.036$	$+0.130 \pm 0.030$	0.47 / 0.58	0.461	0	4
GA 20	$+0.144 \pm 0.033$	$+0.116 \pm 0.020$	0.36 / 0.54	0.109	0	2
GA 25	$+0.072 \pm 0.026$	$+0.074 \pm 0.024$	0.19 / 0.29	0.461	2	0
NPO 10	$+0.283 \pm 0.040$	$+0.187 \pm 0.032$	0.75 / 0.86	0.008	0	2
NPO 25	$+0.166 \pm 0.032$	$+0.148 \pm 0.026$	0.43 / 0.68	0.008	0	0
NPO 40	$+0.073 \pm 0.023$	$+0.110 \pm 0.021$	0.18 / 0.53	0.008	1	0
NPO 55	$+0.039 \pm 0.020$	$+0.038 \pm 0.017$	0.10 / 0.14	0.641	1	0
NPO 75	$+0.008 \pm 0.012$	$+0.029 \pm 0.020$	0.03 / 0.11	0.383	0	1

cells, Wilcoxon $p = 2 \cdot 10^{-7}$) and 0.21 ± 0.04 for Pythia-1B (positive in 32 of 40, $p = 6 \cdot 10^{-7}$). Before unlearning the paraphrase advantage is 57 to 60% of the literal one at both sizes. The effect is largest under NPO: after 40 steps on Pythia-1B the literal score retains 18% of its advantage and the paraphrase score 53%, so the literal score understates what survives by a factor of three. Under GA on Pythia-1B the per-strength differences are smaller, and only the lightest setting is significant on its own.

The larger model is harder to unlearn. At every strength Pythia-1B retains more of the literal advantage than Pythia-410M (for example 0.43 against 0.27 after 25 NPO steps), so the strengths at which the literal advantage is small while the paraphrase advantage is not move towards stronger unlearning: NPO 25–40 for Pythia-410M and NPO 40–55 for Pythia-1B. A verifier does not know which strength a provider used, which argues for scoring paraphrases routinely rather than tuning the audit to one setting.

Table 5 Fraction of runs (seeds × 40 reveal orders) ending REVOKED under three audit rules, at the strengths where they differ; $\varepsilon = 0.2$, $\alpha = 0.05$. The last column is the fraction of runs in which the literal audit issues a certificate and the union audit revokes.

Model	Subject	literal	union	mixture	lit. issued, union revoked
410M	GA 15	0.65	0.84	0.90	0.20
410M	GA 20	0.40	0.58	0.68	0.19
410M	NPO 25	0.22	0.57	0.54	0.38
410M	NPO 40	0.05	0.19	0.20	0.17
1B	GA 15	0.63	0.66	0.65	0.04
1B	GA 20	0.48	0.45	0.46	0.02
1B	NPO 40	0.20	0.40	0.42	0.22
1B	NPO 55	0.01	0.12	0.10	0.11

Auditing paraphrases changes verdicts where it matters, and only there. Single-run verdict counts are noisy here, because two tests near their thresholds disagree in both directions (columns A and B of Table 4). To compare audit *rules* rather than single runs, we re-run the verifier offline on the stored score differences of every cell with 40 reveal orders. Three rules are compared at $\varepsilon = 0.2$, $\alpha = 0.05$: the literal score alone; a *union* of two separate audits, one on the literal score and one on the two calibrated paraphrase scores, each at level $\alpha/2$, which revokes if either revokes and issues only if both issue, so that its false-alarm probability under the exact null is at most α; and VOUCH's *mixture* engine over all six scores at level α.

Table 5 shows the result. With no or light unlearning (none, GA 5, NPO 10) all three rules revoke in at least 97% of runs at both sizes, and with strong unlearning (GA $\geq$ 50, NPO $\geq$ 75) all three revoke in at most 3.3%. In between, the union and the mixture revoke far more often than the literal score alone: after 25 NPO steps on Pythia-410M, 38% of runs would issue a literal-only certificate that the union audit revokes, and after 40 NPO steps on Pythia-1B, 22%. Under GA on Pythia-1B the rules do not differ, consistent with its small retention gap. The paraphrase scores are weaker on their own: on the positive control the paraphrase audit alone revokes in 7 of 8 seeds at both sizes. They therefore belong next to the literal score in the declared class, not in its place, and adding them costs a few forward passes per pair.

7.5 Comparison with membership-inference audits

PLACEHOLDER

8 Discussion and limitations

What the extension changes in practice.

Three recommendations follow for a verifier who audits with VOUCH. First, canaries should be drawn from the distribution of the records they stand in for, because every design that departs from it visibly is detected with advantage one and leaves the

organic claim of Corollary 4 empty. Second, when the facts behind the canaries can be paraphrased, the declared class should contain both the literal score and paraphrase scores calibrated by the base model, audited as a union at level $\alpha/2$ each: under partial unlearning the union revokes runs that a literal-only audit would certify, and elsewhere it agrees with it. Third, calibrating paraphrase scores by the base model costs nothing and matters when the base model has strong priors between values: on the GPT-2 pilot it raised the paraphrase advantage of the positive control from $+0.02$ to $+0.17$, while on Pythia-410M it changed the advantage by less than 0.07 in every seed.

The distinguishing advantage cannot be certified.

Theorem 3 is conditional on γ, and data bound γ only from below. Our detectors are two cheap families; a provider with more compute could train stronger ones. What the theorem does establish is the direction of the burden. A provider who passes the audit on canaries while leaving organic data untouched has, in effect, built a canary detector, and the size of the gap it obtains is bounded by that detector's advantage. Closing the conditional requires either a computational assumption on the provider or a cryptographic attestation that the committed pipeline was run, which VOUCH composes with [26].

Matching real data is the hard part.

Organic-matched twins reach chance-level detectability here because we wrote the organic generator. In a deployment the organic distribution is the users' own data, which the verifier sees only through samples, and a canary generator fitted to those samples inherits their idiosyncrasies and their privacy constraints. Generating matched canaries for real corpora, and measuring their detectability against detectors trained by an adversary with access to the corpus, is the main open problem this work leaves.

Scope of the experiments.

All facts are fictitious and all records are short biographies from one domain. Fictitious facts are a deliberate choice, since they guarantee that the base model knows neither value, but whether the retention gap persists for facts the model also learned in pre-training is untested. The models have at most one billion parameters and are fine-tuned with LoRA adapters; Pythia-1B already behaves differently from Pythia-410M in how much unlearning it needs, so the strengths at which the gap is largest should be expected to move with scale. The forget sets of the partial-unlearning experiment are constructed to contain only the literal form, which is the setting in which squeezing is expected; when the forget set contains every trained form, GA and NPO remove both advantages (Table 3).

The theory.

Theorem 2 conditions on the training randomness and assumes bounded influence, which holds with $\kappa = \ell = 1$ under exact unlearning but is not verified for partial unlearning. The organic advantage Δ^o_ξ is a counterfactual quantity: organic records were not included by a coin, so it is the advantage the pipeline would show on organic data under randomised inclusion, the standard definition of membership leakage, not

an observed rate. Theorem 6 is conservative by a factor of two to four and covers a constant-stake mixture rather than the adaptive experts of the VOUCH engine.

9 Conclusion

VOUCH certifies forgetting on a committed cohort of planted canaries. This paper asked what such a certificate says about the data users delete, and what it can see. Semantic twins keep the exact null, so canaries can carry natural-language facts that have paraphrases. The certificate transfers to the generator population at the cost of a McDiarmid term, and from canaries to organic records at the cost of twice the provider's ability to tell them apart; existing canary designs make that cost maximal and organic-matched twins bring it to chance level against the detectors we built. Under partial unlearning the trained form disappears faster than the fact at both model sizes we tested, and an audit that also scores paraphrases revokes runs that a literal-only audit would certify. The certificate remains a statement about a declared score class, a committed cohort and a provider whose pipeline is attested or trusted; within those limits, it now reaches further towards the claim a deletion request actually asks for.

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- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable. The work uses no human participants and no personal data; all records in the experiments are fictitious and generated by the released code.
- **Consent for publication.** Not applicable.
- **Data availability.** All data are generated deterministically from committed seeds by the released code. Every result reported in this paper, including all per-seed records, is available in the repository listed below.
- **Materials availability.** Not applicable.
- **Code availability.** The code is available at https://github.com/vinhqdang/VOUCH-Verifiable-Online-Unlearning-Certification-via-Hypothesis-betting, in the directory `extension/`.
- **Author contribution.** Q.-V. D. conceived the work, developed the theory, implemented and ran the experiments, and wrote the manuscript.

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